

①

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Assignment 2
121 535 77

1a) $A^3 - 3A^2 + 3A - I = 0$

$$A^3 - 3A^2 + 3A = I$$

$$A(A^2 - 3A + 3I) = I$$

$$A^2 - 3A + 3I = I A^{-1}$$

$$A^{-1} = A^2 - 3A + 3I$$

 $\therefore A$ is invertible

b) $A^{-1} = A^2 - 3A + 3I$

c) $A^3 - 3A^2 + 3A - I = 0$

$$A(A^3 - 3A^2 + 3A - I) = 0$$

$$A^4 - 3A^3 + 3A^2 - A = 0$$

$$A^4 = 3A^3 - 3A^2 + A$$

2a) To find the determinant of A let $x=0$

$$\therefore p(0) = \det(A - 0I) = \det(A)$$

$$p(0) = -(0)^3 + 7(0)^2 - 5(0) + 8$$

$$\det(A) = 8$$

b) $-A^3 + 2A^2 - 5A + 8I = 0$

$$A^3 - 2A^2 + 5A = 8I$$

$$\frac{1}{8}(A^3 - 2A^2 + 5A) = 8I$$

$$\frac{1}{8}(A^2 - 2A + 5)A = I$$

$$\frac{1}{8}(A^2 - 2A + 5)AA^{-1} = IA^{-1}$$

$$A^{-1} = \frac{1}{8}(A^2 - 2A + 5)$$

 $\therefore A$ is invertible

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$$c) \rho(x) = -x^3 + 2x^2 - 5x + 8$$

$$\text{trace} = 2 \implies$$

$$3) (I - BA)^{-1} = I + B(I - AB)^{-1}A$$

$$\text{let } (I - BA) = X$$

$$\text{and } (I - AB) = Y$$

$$\therefore X^{-1} = I + BY^{-1}A$$

$$X^{-1} - I = BY^{-1}A$$

$$B^{-1}(X^{-1} - I) = B^{-1}BY^{-1}A$$

$$B^{-1}X^{-1} - B^{-1} = Y^{-1}A$$

$$(B^{-1}X^{-1} - B^{-1})A^{-1} = Y^{-1}AA^{-1}$$

$$B^{-1}X^{-1}A^{-1} - B^{-1}A^{-1} = Y^{-1}$$

$$\therefore (I - AB)^{-1} = B^{-1}(I - BA)^{-1}A^{-1} - B^{-1}A^{-1}$$